

ISHIM, Russian Federation

WMO: 285730

Lat: 56.100N

Lon: 69.433E

Elev: 270

StdP: 14.55

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.7	-24.0	-34.5	0.8	-29.6	-28.9	1.1	-23.8	19.6	22.0	18.2	19.8	2.2	200	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	85.4	67.8	82.4	66.4	79.2	64.9	70.2	81.5	68.5	78.8	66.8	76.3	7.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.9	74.9	64.7	92.5	73.6	63.0	87.1	72.2	34.4	81.6	32.9	78.8	31.6	76.6	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.7	15.0	-34.9	90.2	6.8	2.9	-39.8	92.3	-43.8	94.0	-47.6	95.6	-52.5	97.7		
(5)			-35.0	73.0	6.7	2.0	-39.9	74.4	-43.8	75.6	-47.6	76.8	-52.5	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	0.6	4.4	19.3	38.2	53.6	63.0	65.3	61.2	50.8	38.3	20.8	7.1		
	DBStd	25.33	13.97	13.34	11.04	10.36	9.44	7.99	6.56	7.26	7.58	9.03	12.84	14.57		
	HDD50	6870	1532	1277	951	372	72	6	0	4	78	374	875	1329		
	HDD65	11034	1997	1697	1416	803	370	130	80	164	430	828	1325	1794		
	CDD50	1538	0	0	0	19	182	398	474	351	103	11	0	0		
	CDD65	226	0	0	0	0	16	71	89	46	4	0	0	0		
	CDH74	2219	0	0	0	8	246	719	785	419	42	0	0	0		
	CDH80	545	0	0	0	1	50	206	181	101	6	0	0	0		
	WSAvg	6.9	6.6	6.9	7.8	8.0	7.7	6.4	5.6	5.7	6.3	6.9	7.5	7.4		
	PrecAvg	14.8	0.6	0.4	0.5	0.9	1.3	2.1	2.7	2.2	1.6	1.1	0.9	0.7		
(15) Precipitation	PrecMax	21.5	1.4	1.3	1.4	2.6	4.6	7.5	4.6	4.5	3.7	4.1	1.9	1.7		
	PrecMin	8.8	0.1	0.0	0.0	0.0	0.1	0.5	0.3	0.0	0.3	0.1	0.2	0.0		
	PrecStd	2.8	0.3	0.3	0.3	0.6	0.9	1.5	1.1	1.1	0.9	0.8	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	35.4	44.7	72.5	85.2	89.3	88.0	86.9	79.1	66.1	45.2	36.0		
		MCWB	31.6	32.4	37.6	55.4	62.4	68.5	69.9	69.1	62.0	52.3	41.8	33.6		
	2%	DB	29.3	31.7	39.4	65.4	80.1	85.5	84.6	82.8	72.8	60.8	41.3	32.9		
		MCWB	27.8	29.8	34.8	50.9	60.2	67.5	69.1	67.1	59.5	49.7	39.2	31.3		
	5%	DB	25.6	28.4	36.7	59.5	76.2	81.9	81.5	78.7	67.6	56.0	38.5	30.0		
		MCWB	24.2	26.4	33.3	48.3	58.5	66.3	67.6	65.5	56.6	47.3	36.1	28.4		
	10%	DB	21.5	24.4	34.4	54.2	71.3	78.0	78.4	74.8	63.6	51.6	35.5	26.7		
		MCWB	20.4	22.8	31.7	45.3	56.3	64.3	66.4	64.0	54.4	45.7	33.6	25.3		
	0.4%	WB	31.7	33.0	38.4	56.6	65.0	71.6	72.8	71.4	64.0	54.3	42.4	33.9		
		MCDB	33.0	34.9	43.7	70.0	79.9	85.3	83.6	83.3	75.5	62.6	44.3	35.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.0	29.8	35.4	51.9	61.9	69.4	70.6	69.1	60.1	51.2	39.5	31.4		
		MCDB	29.3	31.5	38.5	62.8	76.5	81.9	81.1	79.3	70.5	58.3	41.6	33.0		
	5%	WB	24.2	26.6	33.7	48.9	59.4	67.4	68.9	67.0	57.9	48.3	36.3	28.6		
		MCDB	25.4	28.1	36.3	59.0	73.9	78.7	78.8	75.9	66.4	54.2	38.1	30.0		
	10%	WB	20.4	22.9	31.8	45.8	57.0	65.4	67.3	64.9	55.3	45.8	33.9	25.3		
		MCDB	21.5	24.4	34.4	53.6	68.8	75.6	76.3	73.3	62.4	51.2	35.3	26.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	17.0	17.2	17.0	20.7	18.7	17.9	18.1	16.9	13.4	10.8	13.4		
		MCDBR	14.3	15.4	15.2	25.6	27.7	24.5	23.0	24.6	23.3	22.0	10.3	12.8		
		MCWBR	13.8	14.2	11.7	14.4	12.6	10.5	10.0	11.6	11.9	12.7	8.9	12.2		
		MCDBR	13.9	16.2	13.6	24.0	25.5	22.1	20.5	21.8	22.0	19.4	10.3	12.3		
	5% WB	MCWBR	13.5	15.1	11.2	14.2	12.7	10.1	9.8	11.1	12.1	12.0	9.3	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.313	0.369	0.417	0.408	0.408	0.425	0.434	0.381	0.357	0.316	0.282		
	taud		2.043	2.022	1.939	2.064	2.225	2.289	2.239	2.180	2.290	2.213	2.094	1.996		
	Ebn at Noon		186	227	242	251	263	265	257	244	242	212	168	147		
	Edh at Noon		20	31	44	45	41	39	40	40	31	26	19	16		
(44) All-Sky Solar Radiation	RadAvg		266	593	1085	1435	1714	1844	1749	1364	909	459	218	163		
	RadStd		20	35	71	124	123	148	149	134	89	75	36	23		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page